

KARL HILL

MISSION SOFTWARE

ENGINEERING SYSTEMS | TECHNICAL LEADERSHIP | DELIVERY

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SUMMARY

Staff Aerospace Software Engineer with 20+ years of experience building and delivering production software across national security, aerospace, NASA, and enterprise environments. Combines hands-on engineering with technical leadership, leading software delivery, engineering practices, mentoring, and cross-team execution for mission-critical systems.

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 — PRESENT

Jacobs — National Security · Chantilly, VA

- Lead engineering delivery for a ~10-engineer team across ~20 repositories and multiple deployment environments, sequencing work, coordinating dependencies, and driving integration and release readiness.
- Develop mission software, shared interfaces, messaging integrations, and service orchestration while providing technical guidance and reviewing implementation across the team.
- Establish engineering guardrails through CI/CD, automated testing, security checks, repository standards, dependency management, and release automation.
- Lead Agile planning and execution across internal and partner teams, translating mission priorities into sequenced engineering work and resolving cross-team dependencies and blockers.
- Onboarded and coached approximately six engineers through code review, technical feedback, development guidance, and reinforcement of engineering standards.

Lead Software Engineer

DEC 2017 — SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Led design and development of an AWS-based flood-mapping system, automating processing and distribution of satellite-derived flood products.
- Modernized LAADS DAAC web systems, including Find Data search and ordering and near-real-time access, with GitLab CI/CD and Kubernetes delivery.
- Modernized NASA Earth Observatory's web platform, improving performance, user experience, and search discoverability for an audience of approximately 1.5 million monthly visitors during that work.
- Automated a content-registry workflow, reducing manual dataset-registration work by roughly 60% and improving researcher access.
- Built a Ceph-based file and metadata platform for virtual directory mapping and discovery of large scientific datasets.
- Led Agile software delivery across NASA teams, coordinating priorities, sprint execution, stakeholder needs, and continuous improvement across Earth science projects.

DETAILS

Washington, DC

(202) 599-1442

karlhillx@gmail.com

LINKS

LinkedIn

linkedin.com/in/khill

GitHub

github.com/karlhillx

Website

karlhill.com

AREAS OF EXPERTISE

- Software Engineering
- Technical Leadership & Direction
- Distributed Systems & Integration
- Engineering Systems & Developer Tooling
- Engineer Development & Coaching
- Agile & Cross-Team Delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 — DEC 2017

InformedDNA · Washington, DC

- Designed and delivered a Laravel case-management platform connecting counseling workflows, documentation, and billing.
- Built CRM enhancements for customer lifecycle workflows and reporting, and tightened maintenance and security processes with operations.

Senior Software Engineer

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, DC

- Delivered SugarCRM solutions for more than 20 clients, including the Virginia Department of Transportation and Kastle Systems.
- Improved backlog management and delivery practices across client projects.

Earlier Engineering Experience

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal-level roles across travel, enterprise CRM, telecommunications, and managed security.
- Built Java and SQL Server services for Finium, a multi-tenant managed-security platform, and contributed to shared testing and code-quality practices.

AREAS OF EXPERTISE

Software Engineering · Technical Leadership & Direction · Distributed Systems & Integration · Engineering Systems & Developer Tooling · Engineer Development & Coaching · Agile & Cross-Team Delivery

EDUCATION

Computer Science coursework, University of Maryland, Baltimore County

Associate of Arts, General Studies — Computer Science Curriculum, Howard Community College

Project Management Certificate, Rutgers University

CERTIFICATIONS

SAFe Agilist, Scaled Agile (in progress)

Professional Scrum Master II, Scrum.org

Professional Scrum Product Owner I, Scrum.org

OPEN SOURCE

bb-run — Run Bitbucket Pipelines locally.

testrisk — Rank the highest-value Python test gaps from coverage, AST, and git.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, Bash

Cloud & Containers: AWS, Docker, Kubernetes, Helm

CI/CD & Developer Tooling: Git, Bitbucket Pipelines, GitLab CI, GitHub Actions, release automation, security scanning

Python Tooling: uv, Ruff, ty, pytest, pre-commit — dependency management, linting, type checking, testing, and automated quality gates

Integration: REST/OpenAPI, RabbitMQ, ActiveMQ, distributed systems

Data: PostgreSQL, MySQL, SQL Server, Elasticsearch/OpenSearch, Redis